



Cafe Sales Analysis using Python, SQL & Matplotlib

Problem Statement

Built an end-to-end sales analysis solution to transform raw café transaction data into meaningful business insights. The project focuses on identifying sales trends, customer purchasing behaviour, product performance, and payment preferences to support data-driven business decisions.

Project Overview

Cleaned, analysed, and visualized café sales data using Python, SQL, and Matplotlib. The project follows a real-world analytics workflow, from data preprocessing to business recommendations, demonstrating practical data analysis skills.

Dataset Information

Attribute	Value
Rows	10,000
Columns	8
Source	Kaggle

Key Features

- Cleaned and validated raw sales data by handling missing values, correcting data types, and removing duplicate records.
- Performed exploratory data analysis (EDA) to uncover sales patterns and customer behaviour.
- Analyzed monthly sales trends and identified peak business periods.
- Evaluated product performance based on quantity sold and revenue generated.
- Compared payment methods to understand customer payment preferences.
- Created clear visualizations to communicate findings and support business decisions.

Tech Stack

- **Python** – Data processing and analysis
- **Pandas** – Data cleaning and manipulation
- **SQLite** – SQL-based business queries
- **Matplotlib** – Data visualization
- **Jupyter Notebook** – Interactive analysis
- **Git & GitHub** – Version control and project management

Business Questions Solved

- Which month generated the highest sales?
- Which products contributed the most revenue?
- Which items sold the highest quantity?
- How are customers choosing to pay?
- Which product performed best each month?
- What business opportunities can be identified from the sales data?

Exploratory Data Analysis (EDA)

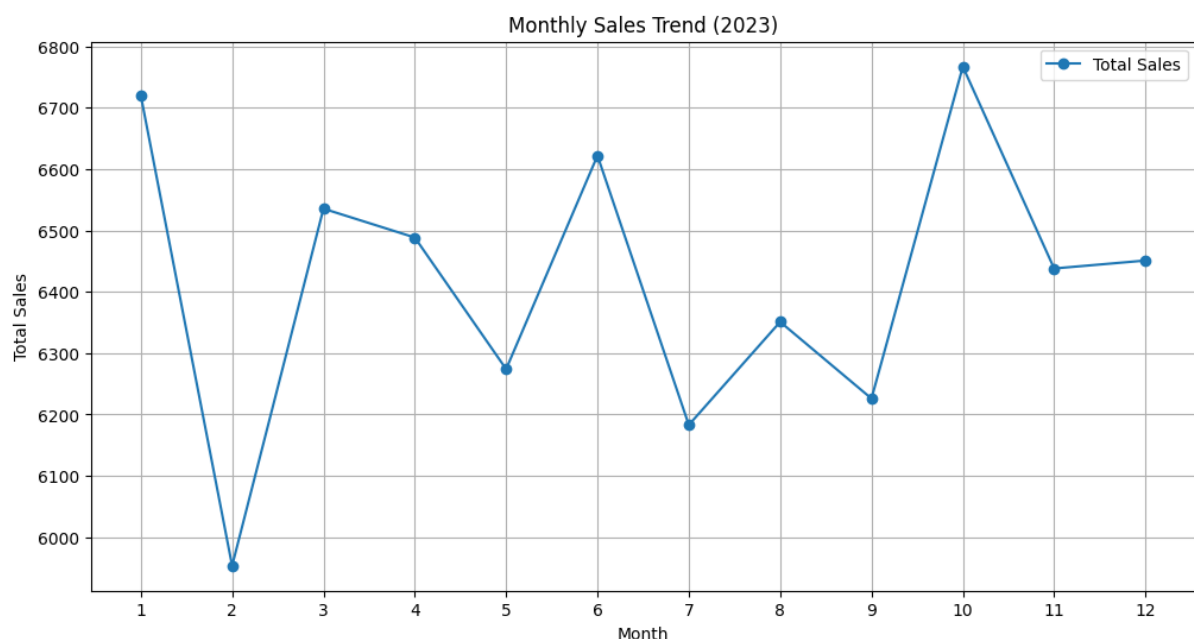
- Monthly sales trend
- Highest sales month
- Best-selling item
- Highest revenue-generating item
- Revenue by payment method
- Most popular payment method
- Monthly top-selling item

Business Impact

Converted raw transaction records into actionable insights that can help optimize inventory planning, identify high-performing products, improve promotional strategies, and better understand customer purchasing behaviour.

Screenshots:

Monthly Sales Trend

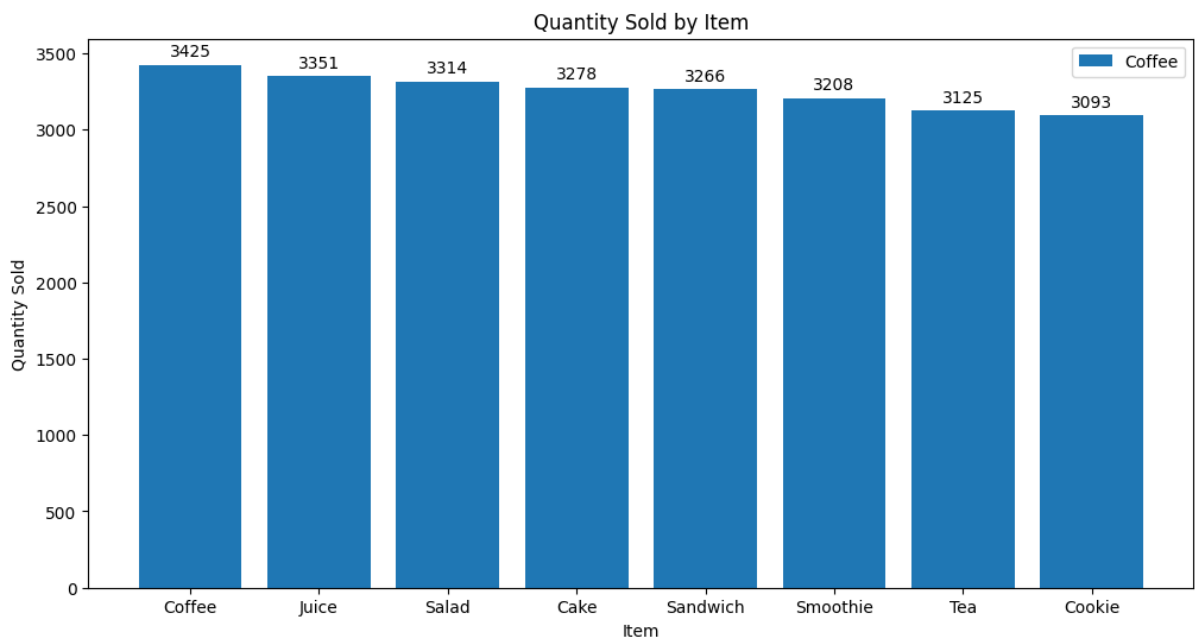


The line chart illustrates the total sales generated each month throughout 2023. Sales remained relatively stable across the year, ranging between approximately ₹5,950 and ₹6,770. October recorded the highest monthly revenue (₹6,767), while February had the lowest (₹5,953.50). Although there were minor month-to-month fluctuations, the overall sales trend remained consistent, indicating steady customer demand.

Business Insight: The café maintained stable revenue throughout the year, with a noticeable sales peak in October. This may be influenced by seasonal demand, festivals, promotions, or increased customer footfall.

Recommendation: Analyze the factors that contributed to October's strong performance and consider replicating similar marketing campaigns or promotional strategies during lower-performing months such as February to improve overall revenue.

Quantity Sold by Item

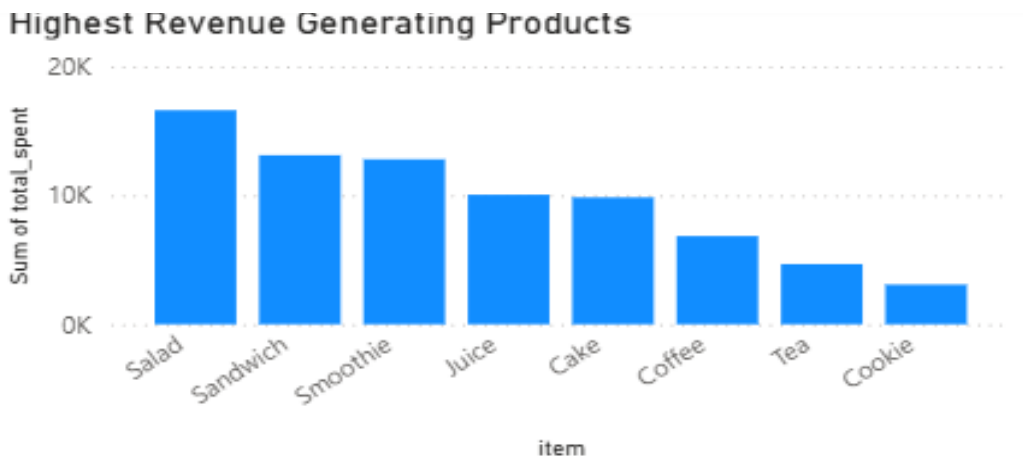


This chart compares the total quantity sold for each menu item during the analysis period. It highlights the products with the highest customer demand and identifies items with comparatively lower sales. Understanding product-wise sales helps evaluate customer preferences and inventory requirements.

Business Insight: The top-selling item reflects the strongest customer demand and should be prioritized for inventory planning and stock availability. Items with lower sales may require promotional offers, menu optimization, or pricing strategies to improve their performance.

Recommendation: Maintain sufficient inventory for high-demand products to avoid stock shortages. Regularly review the performance of low-selling items and consider targeted promotions, combo offers, or menu adjustments to increase customer interest and maximize overall sales.

Highest Revenue Generated by Product

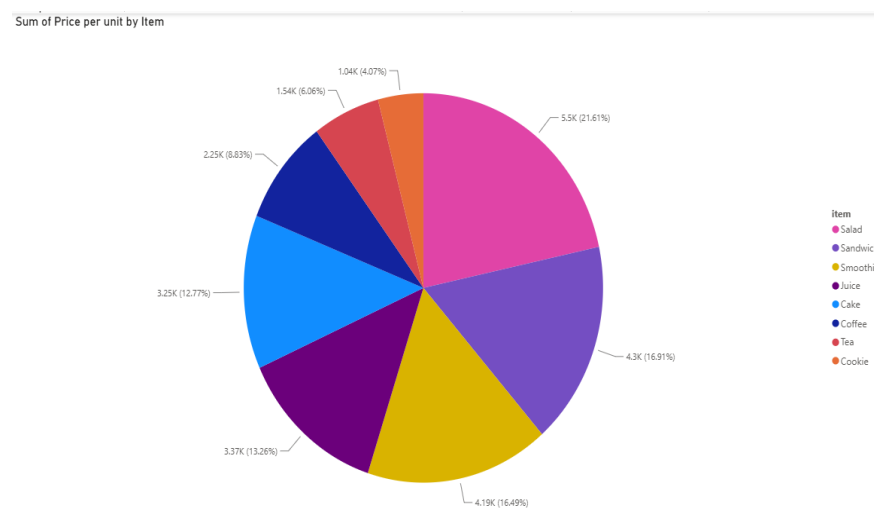


This chart compares the total revenue generated by each product during the analysis period. While some products may sell in high quantities, this analysis highlights which items contribute the most to the café's overall revenue.

Business Insight: Salad generated the highest revenue (₹16,605), followed by Sandwich (₹13,116) and Smoothie (₹12,800). In contrast, Cookie (₹3,082) and Tea (₹4,669.50) contributed the least revenue. This indicates that higher-priced or premium menu items have a significant impact on overall earnings.

Recommendation: Focus on maintaining the availability and quality of high-revenue products such as Salad, Sandwich, and Smoothie. To improve the performance of lower-revenue items, consider introducing promotional offers, meal combos, or pricing strategies that encourage customers to purchase them alongside popular products.

Product price Comparison



This chart compares the cumulative unit prices recorded for each product across all transactions. Products such as Salad, Sandwich, and Smoothie have higher cumulative values due to their higher unit prices and frequency of sales.

Business Insight: Premium-priced products contribute more to the overall pricing value and have the potential to generate higher revenue when customer demand remains strong. However, this metric should be interpreted alongside quantity sold and total revenue for meaningful business decisions.

Recommendation: Focus on revenue and quantity sold as the primary performance indicators. Product pricing analysis should be used to evaluate pricing strategies rather than overall sales performance.

What I Learned

- Built a complete data analysis workflow from raw data to business recommendations.
- Strengthened practical skills in data cleaning, SQL, and exploratory data analysis.
- Improved the ability to communicate technical findings through clear visualizations and business-focused insights.
- Gained experience using Python to solve real-world business problems.

Future Improvements

- Develop an interactive Power BI dashboard for executive reporting.
- Add sales forecasting using machine learning models.
- Build customer segmentation to support targeted marketing strategies.
- Deploy the project as an interactive web application.

How to Run the Project

```
git clone <https://github.com/Ran7714/Cafe_Sales_Analysis>
cd Cafe-Sales-Analysis
pip install -r requirements.txt
jupyter notebook
```

Open: Cafe_Sales_Analysis.ipynb
Run all cells

Project Structure

Cafe-Sales-Analysis/

|

|— data/

| |— cafe_sales_raw.csv

| |— cafe_sales_cleaned.csv

|

|— notebooks/

| |— Cafe_Sales_Analysis.ipynb

|

|— images/

| |— monthly_sales.png

| |— top_selling_items.png

| |— revenue_by_item.png

|

|— reports/

| |— Business_Insights.pdf

|

|— README.md

|— requirements.txt